Weird Algorithm

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1068 | Introductory Problems | https://cses.fi/problemset/task/1068

Consider an algorithm that takes as input a positive integer n. If n is even, the algorithm divides it by two, and if n is odd, the algorithm multiplies it by three and adds one. The algorithm repeats this, until n is one. For example, the sequence for n=3 is as follows: 3 → 10 → 5 → 16 → 8 → 4 → 2 → 1 Your task is to simulate the execution of the algorithm for a given value of n.

Input

The only input line contains an integer n.

Output

Print a line that contains all values of n during the algorithm.

Constraints

- $1 \le n \le 10^6$

Example

Input
3

Output
3 10 5 16 8 4 2 1